

Photochemistry and Phototoxicity of Aloe Emodin[†]

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ABSTRACT

Photochemical pathways leading to the phototoxicity of the aloe vera constituent aloe emodin were studied. The results indicate a photochemical mechanism involving singlet oxygen to be the most likely pathway responsible for the observed phototoxicity. Aloe emodin was found to efficiently generate singlet oxygen when irradiated with UV light ($\Phi_{\Delta} = 0.56$ in acetonitrile). The survival of human skin fibroblast cells in the presence of aloe emodin was found to decrease upon irradiation with UV light. A further decrease in cell survival was observed in D₂O compared with H₂O, suggesting the involvement of singlet oxygen as the primary pathway. Laser flash photolysis experiments were also carried out on aloe emodin alone and in the presence of various biological substrates. Aloe emodin proved to be relatively photostable ($\Phi = 1 \times 10^{-4}$) and a poor photo-oxidant ($E^*_{red} = +1.02$ V). Only absorption bands caused by the triplet state of aloe emodin ($\lambda_{max} = 480$ nm) and the aloe emodin conjugate base ($\lambda_{max} = 520$ nm) were observed in the transient spectra.

INTRODUCTION

The phototoxicity of cosmetic ingredients is a concern because of the large number of products applied to sun-exposed skin. Indeed, several cosmetic ingredients have already been banned because of their enhancement of the sun's damage to skin (1). In this study photochemical mechanisms that may cause the phototoxicity of aloe emodin, a constituent of aloe vera, were investigated. Aloe emodin is found both free and as a glycoside in plants having increasing commercial importance, including aloe and senna (2). The use of aloe alone as a dietary supplement and cosmetic ingredient has been estimated to exceed \$100 billion in worldwide retail sales (The International Aloe Science Council, personal communication). A recent study by Strickland *et al.* (3) found that aloe emodin placed on the skin of

mice along with exposure to UVB radiation resulted in the development of melanomas. However, the biochemical reactions responsible for initiating photocarcinogenesis are unknown. Despite the potential for significant exposure to aloe emodin, very little is known about its photophysical and photochemical properties. Aloe emodin is an anthraquinone derivative similar in structure to the known photodynamic sensitizer hypericin (Fig. 1). Hypericin photosensitization produces both singlet oxygen, which can oxidize cellular substrates, and free radicals, which are produced by Type-I photosensitization (4).

The following study considered three primary photochemical pathways that could be responsible for the phototoxicity and photocarcinogenicity of aloe emodin. The first mechanism entails the direct decomposition of aloe emodin to form reactive intermediates upon photoexcitation with UV light. The second mechanism involves the decomposition of aloe emodin in the presence of both light and a biological substrate. Both these mechanisms were studied by laser flash photolysis (LFP), which was used to monitor the formation of any reactive products upon excitation of aloe emodin. The third mechanism involves absorption of a photon of energy by aloe emodin followed by energy transfer to oxygen to form singlet oxygen. Singlet oxygen can then react with intracellular targets, such as DNA and cell membranes, to elicit a phototoxic response (5). The production of singlet oxygen was determined by direct detection of the emission from singlet oxygen at 1270 nm. Finally, the binding of aloe emodin to DNA and the photocleavage of plasmid DNA sensitized by aloe emodin were investigated. The results of these experiments show aloe emodin to be relatively photostable when irradiated with UV light both in the presence and in the absence of biological substrates. However, aloe emodin was found to efficiently generate singlet oxygen leading to decreased cell survival.

MATERIALS AND METHODS

Chemicals. Aloe emodin and calf thymus DNA were obtained from (Sigma Chemical Co., St. Louis, MO) and used without further purification. Rose bengal was obtained from (Aldrich, Milwaukee, WI) and used as received. All solvents were of spectroscopic grade and were used without purification. 1,8-Dimethoxy-3-(hydroxymethyl)-9,10-anthraquinone was prepared according to the procedure described by Alexander *et al.* (6), and its ¹H NMR was consistent with the literature spectrum.

General techniques. Ground-state absorption spectra were obtained using a Perkin-Elmer Lambda 2S spectrometer. Fluorescence emission spectra were obtained using a Gilford Fluoro IV spectro-

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Abbreviations: LFP, laser flash photolysis; MeCN, acetonitrile; NHE, normal hydrogen electrode; PBS, phosphate-buffered saline.

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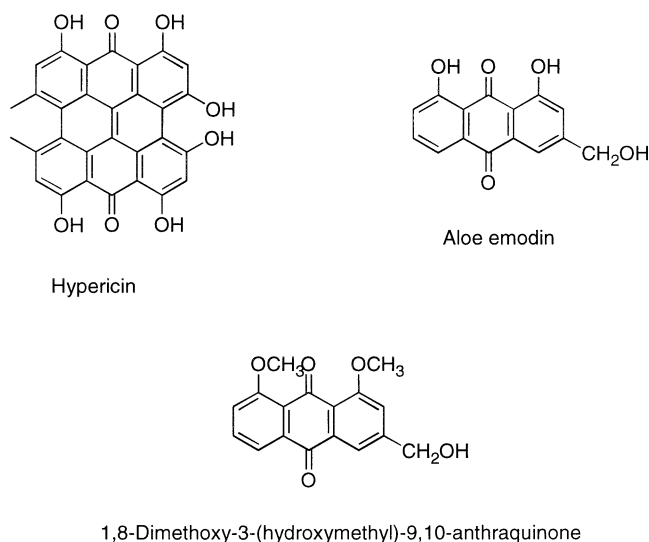


Figure 1. Chemical structures of hypericin, aloe emodin and 1,8-Dimethoxy-3-(hydroxymethyl)-9,10-anthraquinone.

fluorometer. The fluorescence quantum yield of aloe emodin was determined using 9-fluorenone (Aldrich) ($\Phi_f = 0.0027$ in ethanol) as the reference.

LFP experiments. LFP experiments were performed using the third (355 nm) harmonic output from a Continuum II-10 Q-switched Nd:YAG laser as the excitation source. The transient absorption spectra were monitored using a 350 W Xenon lamp, which passes through the sample cell at a right angle to the excitation beam. Transient waveforms were captured using a Lecroy 9420 digital oscilloscope. Samples were prepared with optical densities of 0.20–0.40 at 355 nm and purged for 15 min with N_2 or O_2 for all LFP experiments.

Singlet oxygen quantum yields. Singlet oxygen ($O_2(^1\Delta_g)$) was detected by time-resolved luminescence spectroscopy. Samples were excited using the third harmonic (355 nm, 3 ns) of a Continuum Minilite Nd:YAG laser. A 355 nm band-pass filter was placed at the exit port of the laser to block any residual 1064 nm light. The emission at 1270 nm from $O_2(^1\Delta_g)$ was detected at a right angle to the excitation beam with a liquid N_2 -cooled InGaAs detector. A 1270 nm band-pass filter was placed between the sample cell and the detector to prevent light from the excitation beam from reaching the detector. Experiments were repeated using laser pulse energies ranging from 0.05 to 0.6 mJ/pulse. The singlet oxygen quantum yield was determined by comparing the slope of the plot of phosphorescence intensity versus the laser power for optically matched samples of aloe emodin and the standard sensitizer rose bengal ($\Phi_\Delta = 0.80$ in acetonitrile [MeCN]) (7). Samples were prepared with optical densities of 0.20–0.30 at the excitation wavelength.

Supercoiled plasmid relaxation assay. Supercoiled plasmid pBR322 (Life Technologies, Rockville, MD) was precipitated by addition of ethanol and dissolved in 10 mM phosphate, pH 7.4. The final concentration of the supercoiled plasmid DNA was adjusted to 10 μM in base pairs. Solutions of aloe emodin, anthraquinone-2-sulfonic acid, sodium salt (Aldrich), or rose bengal (Aldrich) were added to yield a final concentration of 20 μM . This concentration was selected for comparison of photosensitizing activities because the limited solubility of aloe emodin in aqueous buffers prevented the preparation of stable solutions above 30 μM . Air-saturated solutions containing plasmid and aloe emodin or anthraquinone-2-sulfonic acid, sodium salt, were then irradiated in 96-well plates with a source of UVA radiation (320–400 nm) consisting of two GE F40BL bulbs filtered through 3 mm of soft glass. The irradiance of this source was typically 2.9×10^{-3} W/cm². The maximal absorption of solutions containing 20 μM aloe emodin or anthraquinone-2-sulfonic acid, sodium salt, in the UVA spectral region was 0.03 or 0.09, respectively. Solutions containing rose bengal were irradiated with visible light provided by two 40 W GE Watt-miser bulbs filtered through Plexiglas® UF3 (Rohm and Haas Co., Bristol, PA). The ir-

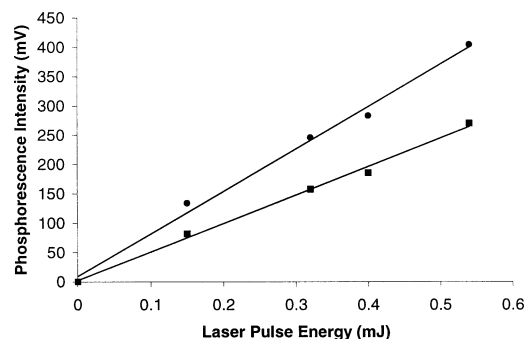


Figure 2. Singlet oxygen phosphorescence intensity versus laser pulse energy for aloe emodin (■) and rose bengal (●) in MeCN.

radiance of this source was typically 2.4×10^{-3} W/cm². The maximum absorption of solutions containing 20 μM rose bengal in the visible spectral region was 1.8. To identify direct photocleavage of DNA, samples were analyzed by electrophoresis on 1.2% agarose gels in tris (hydroxymethyl) aminomethane-borate-ethylenediamine tetraacetic acid disodium salt buffer (TBE) buffer for 2 h at 30 mA. To identify oxidative damage to the purine bases in addition to direct breaks, 25 μL aliquots of irradiated samples were incubated with 0.05 units of *Escherichia coli* formamidopyrimidine-DNA glycosylase (Trevigen, Gaithersburg, MD) for 1 h at 37°C. Cleavage of DNA, reflecting direct strand breaks and enzymatically induced breaks at the sites of oxidized purines, was then assessed by agarose gel electrophoresis. To visualize DNA, gels were stained with ethidium bromide and observed under UV light.

Cell survival. A human skin fibroblast cell line (ATCC CRL 1634) was obtained from the American Type Culture Collection (Manassas, VA). Cells were cultured in Dulbecco modified Eagle medium containing 10% fetal bovine serum (Life Technologies), antibiotics, 4.5 mg/mL glucose and 4 mM L-glutamine. To measure the effects of UVA radiation and aloe emodin on cell survival, subconfluent monolayers of fibroblasts in 60 mm petri dishes were washed twice with phosphate-buffered saline (PBS), pH 7.4, and then treated with PBS containing aloe emodin. A sufficient amount of 20 mM aloe emodin in dimethyl sulfoxide was added to PBS to yield a final concentration of 1 μM aloe emodin. For investigating the effects of D_2O (Aldrich) on photocytotoxicity, additional groups were treated with 1 μM aloe emodin in PBS prepared in D_2O (pD = 7.4). Cells were then exposed to selected fluences of UVA radiation supplied by the filtered fluorescent bulbs described previously. Fibroblasts were maintained at $25 \pm 2^\circ C$ during irradiation, which typically did not exceed 30 min. After irradiation the cells were removed from the substratum by trypsinization, resuspended in medium and seeded into 60 mm petri dishes (500–5000 cells/dish, depending on the expected level of survival). After incubation for 7–10 days at 37°C the cells were fixed with methanol, stained with crystal violet and the cell colonies counted.

RESULTS

Singlet oxygen formation

The ability of aloe emodin to generate singlet oxygen was examined. The quantum yield for singlet oxygen generation by aloe emodin was determined by direct detection of the phosphorescence emission at 1270 nm of $O_2(^1\Delta_g)$. The singlet oxygen quantum yield was determined in MeCN, MeCN/ D_2O mixtures and ethanol. The quantum yield in pure D_2O could not be determined because of the low solubility of aloe emodin in D_2O . Rose bengal was used as the standard singlet oxygen sensitizer in all the solvents. Figure 2 shows the plots of the phosphorescence intensity versus the laser pulse energy for optically matched samples of aloe emodin and the standard rose bengal ($\Phi_\Delta = 0.80$) (7) in MeCN.

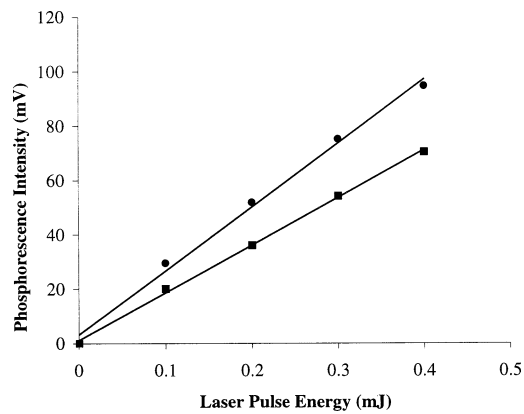


Figure 3. Singlet oxygen phosphorescence intensity versus laser pulse energy for aloe emodin (■) and rose bengal (●) in 50% MeCN/D₂O.

From the ratio of the slopes a quantum yield for singlet oxygen formation of 0.54 was determined for aloe emodin in MeCN. In 50% MeCN/D₂O the singlet oxygen quantum yield for aloe emodin is 0.58 relative to rose bengal (Fig. 3). The quantum yield for singlet oxygen formation did not vary significantly when the fraction of D₂O was increased. A singlet oxygen quantum yield of 0.49 was determined for aloe emodin in ethanol relative to rose bengal ($\Phi_{\Delta} = 0.68$) (8). The significant singlet oxygen quantum yields indicate that singlet oxygen sensitization is a relevant pathway for aloe emodin phototoxicity.

To determine if singlet oxygen formation is relevant to phototoxicity, we examined the effects of UVA radiation and aloe emodin on the survival of skin fibroblast cells. Experiments were carried out in both H₂O and D₂O because the lifetime of singlet oxygen increases from 2 μ s in H₂O to 32 μ s in D₂O (9). Exposure of skin fibroblasts to 1 μ M aloe emodin and UVA radiation results in decreased cell survival (Fig. 4). Treatment of cells with 1 μ M aloe emodin in PBS

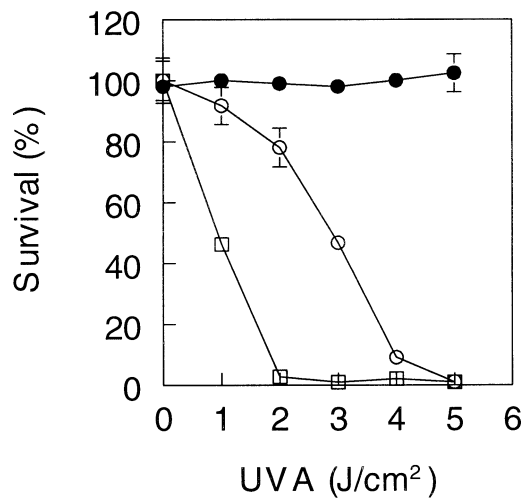


Figure 4. Photocytotoxicity of 1 μ M aloe emodin. Skin fibroblasts were treated with 1 μ M aloe emodin in PBS, prepared in H₂O (○) or D₂O (□), and immediately exposed to the indicated fluences of UVA radiation. Additional groups were treated with UVA radiation only (●). Each data point is the mean $\pm$ standard deviation of four independent determinations.

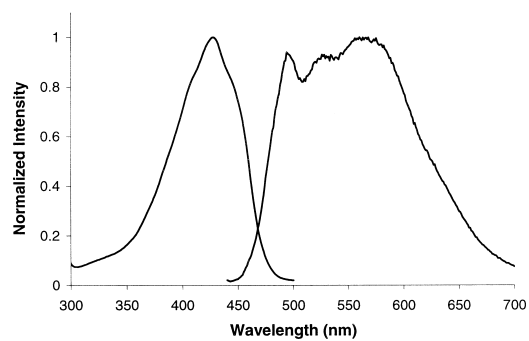


Figure 5. Absorption and emission spectra of aloe emodin in MeCN.

prepared with D₂O results in a further decrease in cell survival. The decrease in cell survival in D₂O relative to H₂O supports a phototoxic mechanism that involves sensitization of singlet oxygen by aloe emodin. Cell survival was unaffected by exposure to UVA radiation alone. Similarly, the survival of cells treated with 1 μ M aloe emodin and maintained in the dark at $25 \pm 2^\circ\text{C}$ was unaffected (data not shown).

Photostability and LFP

Figure 5 shows the absorption and fluorescence emission spectra of aloe emodin in MeCN. The absorption spectrum has a maximum at 430 nm. The fluorescence emission spectrum of aloe emodin excited at 430 nm exhibits structured bands with peaks at 565, 530 and 495 nm. The structured emission spectrum results from contributions from both aloe emodin and the aloe emodin anion to the spectrum. Increasing the fraction of water leads to an increase in the intensity of the 565 and 530 nm peaks relative to the 495 nm peak. The fluorescence quantum yield for aloe emodin was determined to be 0.02 in ethanol. From its zero-zero band the singlet-state energy of aloe emodin can be estimated at 61.1 ± 0.5 kcal/mol (Table 2).

Direct photodecomposition of aloe emodin to form reactive or toxic products is a possible mechanism for its phototoxicity. The photostability of aloe emodin was investigated by irradiating aloe emodin with UV light and monitoring its absorption band at 430 nm as a function of 355 nm light absorbed. This irradiation led to a decrease in absorbance at 430 nm. From this decrease in absorption a quantum yield for photodecomposition of 1×10^{-4} was determined. The small quantum yield indicates that direct decomposition of aloe emodin is not likely to be a significant pathway for its phototoxicity.

Table 1. Quantum yields for singlet oxygen formation for aloe emodin in MeCN and MeCN–D₂O mixtures relative to rose bengal

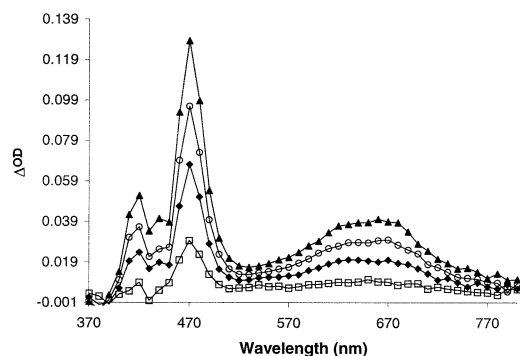
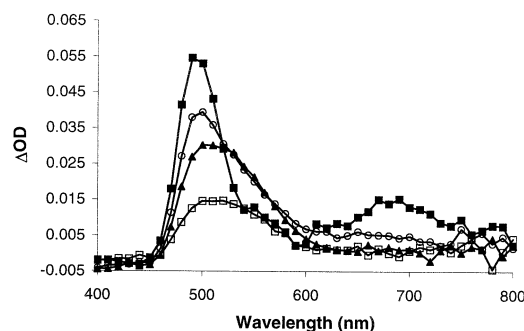
% MeCN	Φ_{Δ}
100	0.54
75	0.53
60	0.60
50	0.58
40	0.49

Table 2. Photophysical parameters, redox potentials and pK_a for aloe emodin

Abs. $\lambda_{\max}$ (nm)	Fl. $\lambda_{\max}$ (nm)	Φ_f	E_S (kcal/mol)	E_T (kcal/mol)	E_{red} (V) versus (NHE)	pK_a
430	565	0.02	61.1 ± 0.5	45.5 ± 3.5	-1.42	9.4 ± 0.2

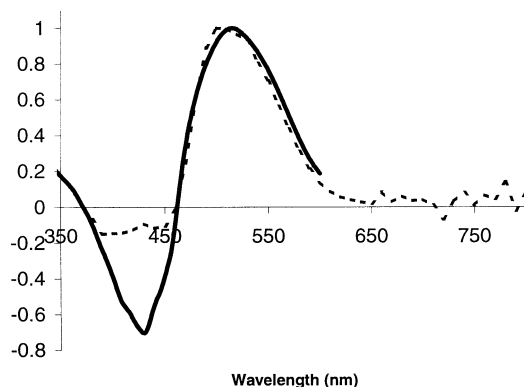
To determine what intermediates are formed upon photolysis, LFP experiments were carried out on aloe emodin in MeCN using 355 nm excitation light. Immediately after the laser pulse, absorption bands are observed at 410, 480 and 670 nm (Fig. 6). These signals decay with rate constants of $3.2 \times 10^5/s$, $3.4 \times 10^5/s$ and $3.4 \times 10^5/s$, respectively, and are attributed to the same species. The absorption bands at 410, 480 and 670 nm are assigned to the triplet state of aloe emodin because they are quenched after purging the solution with oxygen. The triplet state of aloe emodin was also quenched *via* energy transfer by *trans*-stilbene ($E_T = 42$ kcal/mol) and 2,5-dimethyl-2,4-hexadiene ($E_T = 49$ kcal/mol). The quenching rate constants were determined using a pseudo-first-order rate analysis. Quenching by *trans*-stilbene gives a rate constant of $7.7 \times 10^6/(M s)$, whereas quenching by 2,5-dimethyl-2,4-hexadiene is diffusion limited ($7.8 \times 10^9/[M s]$). Triplet energy transfer occurs at the diffusion limit when the energy of the donor exceeds that of the acceptor by *ca* 5 kcal/mol (10). The triplet energy of aloe emodin must therefore lie between that of *trans*-stilbene and 2,5-dimethyl-2,4-hexadiene. We estimate a value of 45.5 ± 3.5 kcal/mol for the triplet-state energy of aloe emodin.

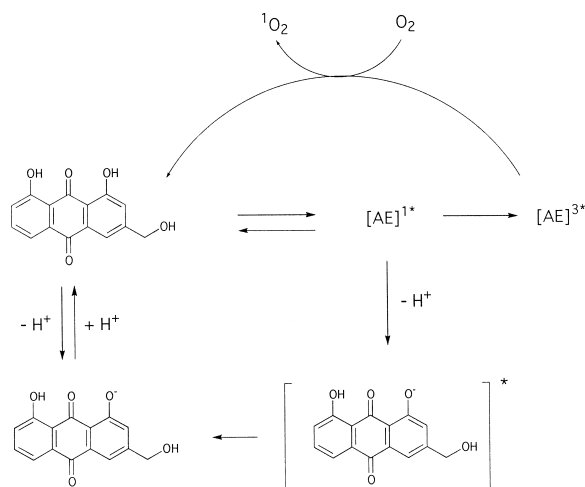
Upon addition of NaOH to a solution of aloe emodin in MeCN, the absorption maximum shifts from 430 to 515 nm. The fluorescence spectrum shows a similar shift of the emission band at 560–650 nm upon addition of NaOH. Addition of acid regenerates the original absorption and fluorescence spectra. The related anthraquinone compound, hypericin, has shown a similar pH-dependent shift in its absorption and emission spectra (11). Multiple pK_a have been reported for hypericin at 11 and 7 (12). A spectrophotometric determination gave a pK_a of 9.4 ± 0.2 for aloe emodin in aqueous solution. A pK_a value of 10.5 has been reported for the first deprotonation of the similar compound, 1,8-dihydroxyanthraquinone (13). A calculation based on the Förster cycle

**Figure 6.** Transient absorption spectra from LFP (355 nm) of aloe emodin in MeCN. (▲) 300 ns, (○) 1 μs , (◆) 2 μs , (□) 5 μs .**Figure 7.** Transient absorption spectra from LFP (355 nm) of aloe emodin in 44% MeCN/H₂O purged with N₂. (■) 100 ns, (○) 1.5 μs , (▲) 5 μs , (□) 20 μs .

estimates a pK_a^* of 1.3 for the excited singlet state of aloe emodin.

LFP experiments on aloe emodin were repeated in a 40% MeCN/H₂O solvent mixture. As shown in Fig. 7, LFP of aloe emodin in this medium gives both the triplet absorption bands as well as an additional longer-lived signal at 520 nm ($\tau = 28 \mu s$). Purging the solution with oxygen quenches the absorption bands resulting from the triplet state of aloe emodin but does not affect the absorption band at 520 nm. This suggests that the species at 520 nm originates from the singlet excited state of aloe emodin and not from its triplet state. When the LFP experiment is carried out at a pH of 3.7, the transient absorption spectrum of aloe emodin in 40% MeCN/H₂O only exhibits an absorption band at 480 nm resulting from the triplet state. This suggests that the absorption band at 520 nm may result from absorption by the aloe emodin anion. This assignment was further examined by comparison of the 520 nm absorption band in the transient spectrum with that in the difference spectrum between the ground-state absorption spectra under neutral and basic conditions (Fig. 8). The good agreement between the transient spectrum and the difference spectrum supports the assignment of the band at 520 nm to the ground-state aloe emodin anion. From these results the proposed mechanism (see Scheme 1) involves excitation of aloe emodin to its first excited state, from where it can then intersystem cross to form the triplet state or lose a proton to form the excited-state aloe emodin anion. The excited-state aloe emodin anion

**Figure 8.** Transient absorption spectrum of aloe emodin (---) versus the steady-state absorption spectrum of aloe emodin anion (—).



Scheme 1.

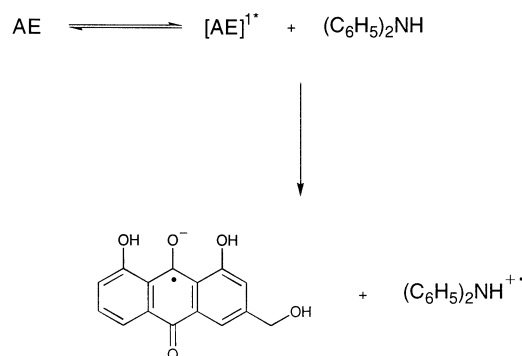
then decays back to the ground state, where it is protonated to give ground-state aloë emodin.

To confirm that the observed signal at 520 nm is caused by the aloë emodin anion, LFP experiments were also carried out on 1,8-dimethoxy-3-(hydroxymethyl)-9,10-anthraquinone (Fig. 1). The absence of hydroxy groups at the 1 and 8 positions on the ring prevents the loss of a proton and the formation of the anion. Figure 9 shows the transient absorption spectra for 1,8-dimethoxy-3-(hydroxymethyl)-9,10-anthraquinone in 40% MeCN/H₂O. An absorption band at 440 nm resulting from the excited triplet state is observed, but no absorption bands at longer wavelengths, equivalent to the transient absorption of aloë emodin at 520 nm, are observed. This result further supports the assignment of the absorption signal at 520 nm to the aloë emodin anion.

Mechanisms for phototoxicity are also known to involve irradiation of the phototoxin in the presence of a biological substrate to form reactive photoproducts through electron transfer mechanisms. For example, it has been demonstrated that some related anthraquinone derivatives can intercalate into and abstract an electron from DNA (14). For this reason we sought to determine if aloë emodin could function as an excited-state electron acceptor. Cyclic voltammetry experiments were carried out on aloë emodin in MeCN. These experiments provide $E_{red} = -1.42$ (versus NHE) for aloë emodin. From Eq. 1 (15) and the singlet energy of aloë emodin we calculate that ΔG for electron transfer will be favorable only when the E_{ox} of a substrate is <1.02 V. Substrates with higher E_{ox} values are predicted to show positive ΔG_{CT} and thus negligible rates of electron transfer to aloë emodin.

$$\Delta G = 23.06 \left[E_{ox} - E_{red} - \frac{e_0^2}{\epsilon R} \right] - E_{00} \quad (1)$$

To investigate the possibility of electron transfer, LFP experiments were carried out on aloë emodin and various electron donors. LFP experiments were carried out where aloë emodin was photolyzed in the presence of tryptophan ($E_{ox} = +1.19$ V) (16) or guanosine monophosphate ($E_{ox} = +1.49$ V) (17) in 40% MeCN/H₂O. As expected from Eq. 1, no evidence of electron transfer was observed, even at high con-



Scheme 2.

centrations of these substrates. The transient absorption spectrum showed only absorption bands resulting from the triplet state and anion of aloë emodin. On the other hand, electron transfer was observed in LFP experiments on aloë emodin and diphenylamine ($E_{ox} = +0.92$ V). An absorption band, which matches the literature value for the diphenylamine radical cation, appears immediately after the laser pulse at 680 nm (18). The signal at 680 nm decays within 2 μ s, and a new signal corresponding to the literature value for the neutral diphenylamine radical appears at 710 nm (19). A new signal is also seen at 450 nm 1.5 μ s after the laser pulse. Addition of diphenylamine also quenches the steady-state fluorescence spectrum of aloë emodin with a $k_q\tau$ value of 20.9/M. This provides further evidence for the electron transfer occurring in the excited singlet state of aloë emodin.

DNA binding and photocleavage

The binding constant of aloë emodin to DNA was determined by monitoring the steady-state fluorescence of aloë emodin as a function of DNA base pair concentration. Addition of DNA to a solution of aloë emodin results in a decrease in the fluorescence intensity of aloë emodin at 560 nm. Calf thymus DNA was added to a 3 μ M solution of aloë emodin containing 50 mM Tris buffer, 1 mM ethylenediaminetetraacetic acid and 50 mM NaCl. The concentration of the DNA base pairs used ranged from 19.8 to 194.3 μ M. The fraction of aloë emodin bound to DNA (α) was determined from Eq. 2, where I_1 is the fluorescence intensity of free aloë emodin with no DNA present, I_2 is the fluorescence

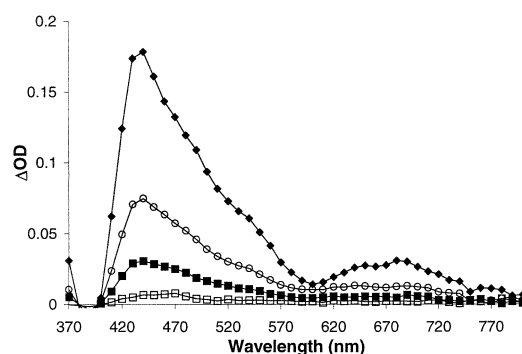


Figure 9. Transient absorption spectra of methylated aloë emodin in 40% MeCN/H₂O. (◆) 100 ns, (○) 300 ns, (■) 500 ns, (□) 1 μ s.

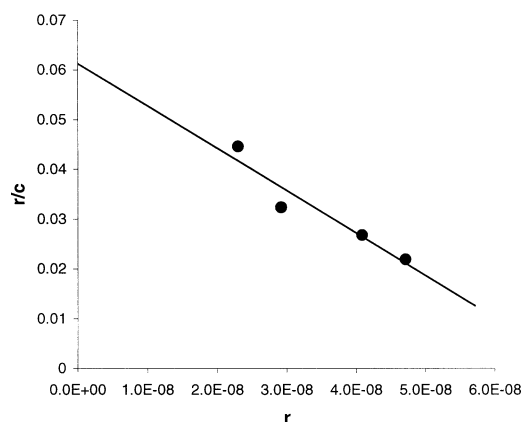


Figure 10. Scatchard plot for aloe emodin binding to DNA.

intensity when all aloe emodin is bound, and I is the fluorescence intensity of a mixture of free and bound aloe emodin. The binding constant (K_b) was then determined from Eq. 3, where r is the amount of bound aloe emodin per DNA base pair, c is the concentration of free aloe emodin, and n is the number of binding sites per base pair. A Scatchard plot of r/c versus r yields a straight line

$$\alpha = (I_1 - I)/(I - I_2) \quad (2)$$

$$\frac{r}{c} = K_b n - K_b r \quad (3)$$

with a slope equal to $-K_b$ (Fig. 10). From this plot a binding constant for aloe emodin and DNA of $8.5 \pm 1.5 \times 10^5/M$ was determined. For comparison, the well-known DNA intercalator ethidium bromide has a binding constant of $2.6 \times 10^6/M$ (20).

Identification of cellular targets that are damaged following photoexcitation of aloe emodin is critically important for understanding the photobiological activity of aloe emodin. The results given in Fig. 11A indicate that direct damage to DNA, which would presumably proceed through electron transfer from DNA to aloe emodin, does not occur. This result is consistent with our cyclic voltammetry experiments, which indicate that aloe emodin is a relatively poor excited-state electron acceptor. In contrast, anthraquinone-2-sulfonic acid, which has been shown to be a good excited-state electron acceptor (20), readily photosensitizes the nicking of plasmid DNA (Fig. 11A). Additionally, rose bengal photosensitized the nicking of plasmid DNA, as previously demonstrated by other investigators (21,22). In addition, consistent with previous reports (20,21), we found that anthraquinone-2-sulfonic acid and rose bengal photosensitize oxidative damage, detectable after digestion of plasmid DNA with *E. coli* formamidopyrimidine-DNA glycosylase (Fig. 11B). The results, given in Fig. 11B, also demonstrate that aloe emodin does not photosensitize the oxidative damage to purine bases in plasmid DNA. This result is particularly interesting because in LPF studies we have determined that aloe emodin efficiently photosensitizes the formation of singlet oxygen (*vide ante*). It should be noted that experimental conditions in LPF studies and plasmid nicking experiments significantly differ. The concentrations of aloe emodin used in LPF studies were 5–10 times higher than the concentration

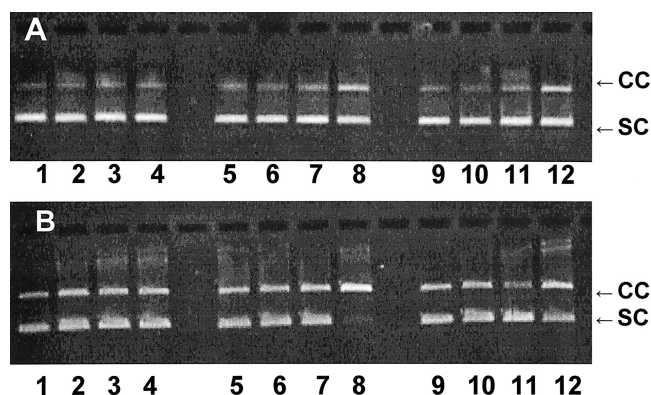


Figure 11. A: Photocleavage of plasmid DNA sensitized by aloe emodin (AE), anthraquinone-2-sulfonic acid (AS) and rose bengal (RB). Samples were analyzed by electrophoresis on 1.2% agarose gels in TBE buffer for 2 h at 30 mA, stained with ethidium bromide and visualized under UV light. Lanes 1, 5 and 9 are untreated plasmid DNA. Lanes 2 and 6 are plasmid DNA irradiated with 2.5 J/cm^2 UVA. Lane 10 is plasmid DNA irradiated with 2.5 J/cm^2 visible light. Lanes 3, 7 and 11 are plasmid DNA treated with $20 \mu\text{M}$ AE, AS or RB, respectively. Lanes 4 and 8 are plasmid DNA treated with $20 \mu\text{M}$ AE or AS, respectively, and then irradiated with 2.5 J/cm^2 UVA. Lane 12 is plasmid DNA treated with $20 \mu\text{M}$ RB and 2.5 J/cm^2 visible light. B: Photosensitized damage to purine bases in plasmid DNA. Samples were treated as in Fig. 1A. After irradiation the samples were digested with 0.05 units of *E. coli* formamidopyrimidine-DNA glycosylase and analyzed by agarose gel electrophoresis. SC is the supercoiled form of plasmid DNA, and CC is the relaxed closed circular form of plasmid DNA, resulting from single- or multiple-strand breaks.

used in experiments done to determine the ability of aloe emodin to photosensitize the nicking of plasmid DNA. In addition, the intensity and spectral distribution of UV light differed in these experiments. Our combined results suggest that although aloe emodin photosensitizes the formation of singlet oxygen, the levels of singlet oxygen produced under conditions of biologically relevant concentrations are insufficient to produce the nicking of plasmid DNA. However, we cannot eliminate the possibility that other indirect pathways leading to DNA damage may exist in a cellular environment. Additional studies are needed to definitely determine the role of DNA damage in the phototoxicity elicited by aloe emodin.

DISCUSSION

Our results demonstrate that the exposure of human skin fibroblasts to aloe emodin and UV radiation elicits significant phototoxicity. Although *in vitro* studies on the toxicity (23) and genotoxicity (24) of aloe emodin and its effects on the promotion of malignant transformation (25) have been reported, the phototoxicity of aloe emodin has not been previously established. However, several anthraquinones, structurally similar to aloe emodin, have been reported to be phototoxic. Emodin, a naturally occurring structural isomer of aloe emodin, has been found to be phototoxic to Chinese hamster V79 cells (26). In addition, synthetic, diamino-substituted anthraquinones developed as antitumor agents have been shown to be phototoxic to leukemic cells (27). Hartely *et al.* (27) found that 1,5- and 1,8-disubstituted aminoanthraquinones are phototoxic to human leukemic cells,

whereas 1,4-disubstituted analogs are inactive (27). The ability to photosensitize cellular toxicity was reported to reflect the photosensitizer's efficiency for intersystem crossing (27). These investigators also determined that phototoxicity was accompanied by the appearance of single-strand breaks in cellular DNA (27).

Although the detailed mechanism underlying the photobiological activity of aloe emodin remains to be established, the results of this study clearly define approaches for further photobiological studies. Our results indicate a photochemical mechanism involving singlet oxygen to be the most likely pathway responsible for the phototoxicity of aloe emodin. Aloe emodin was found to efficiently generate singlet oxygen upon irradiation with UV light. Phototoxicity studies on human skin fibroblast cells in the presence of aloe emodin showed a decrease in cell survival in D₂O compared with H₂O, supporting a singlet oxygen pathway. These results indicate that future studies of the photobiological properties of aloe emodin should include assessments of photooxidative damage to cellular targets. Aloe emodin was also found to be photostable and a poor direct photooxidant because of its relatively large negative reduction potential. Finally, although aloe emodin was found to bind to DNA, single-strand breaks in plasmid DNA photosensitized by aloe emodin were not observed. Whereas these results suggest that photoexcitation of aloe emodin does not lead directly to DNA damage, indirect pathways involving intermediate photooxidation of other cellular components may lead to cellular DNA damage. Additional studies are currently in progress to identify the cellular targets damaged following photoexcitation of aloe emodin.

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